

Final Model Pictures

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Table of Contents:

Computerized Receptor Visualization - Page 1

Initial 3D Printed Model - Page 6

Final 3D Printed Model - Page 8

3D Visualizations of Inactive vs. Active D3R Conformational Changes:

Video:

https://drive.google.com/file/d/17DK6lpzOJtSpvtC7GQ1NMs1xgNSPA5d3/view?usp=drive_link

Images:

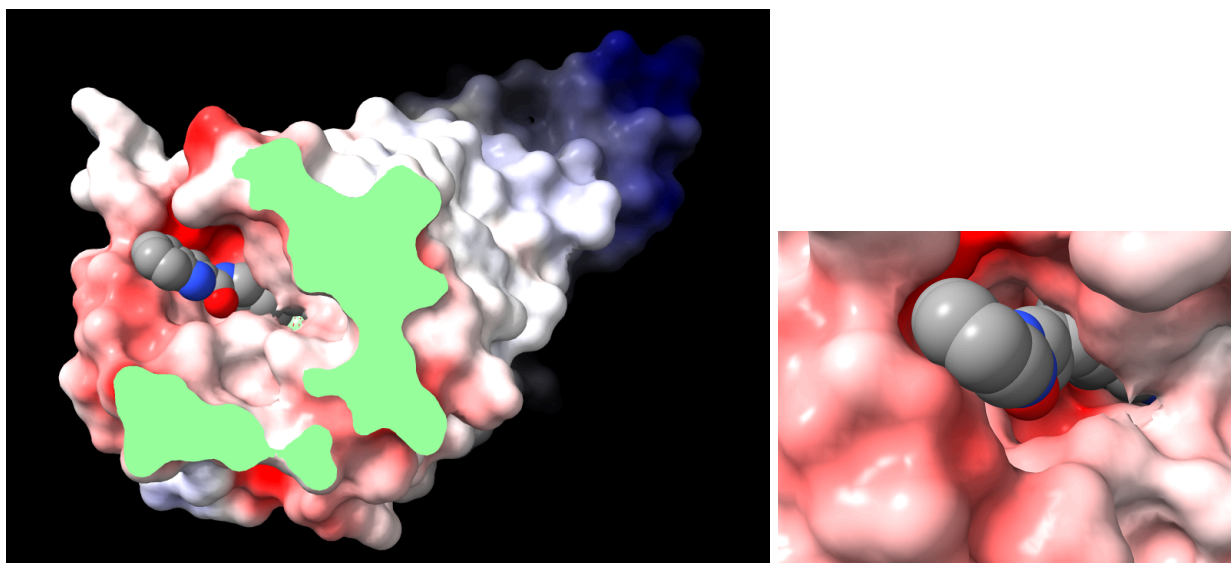


Fig 1. Extracellular view of active D3R with surface electrostatic coloring (red: areas with high electron density/negative potential, attracting positive ions; blue: areas with low electron density/positive potential, attracting negative ions) and the bitopic ligand in atomic sphere form, alongside a closeup to visualize how the two molecules interact.

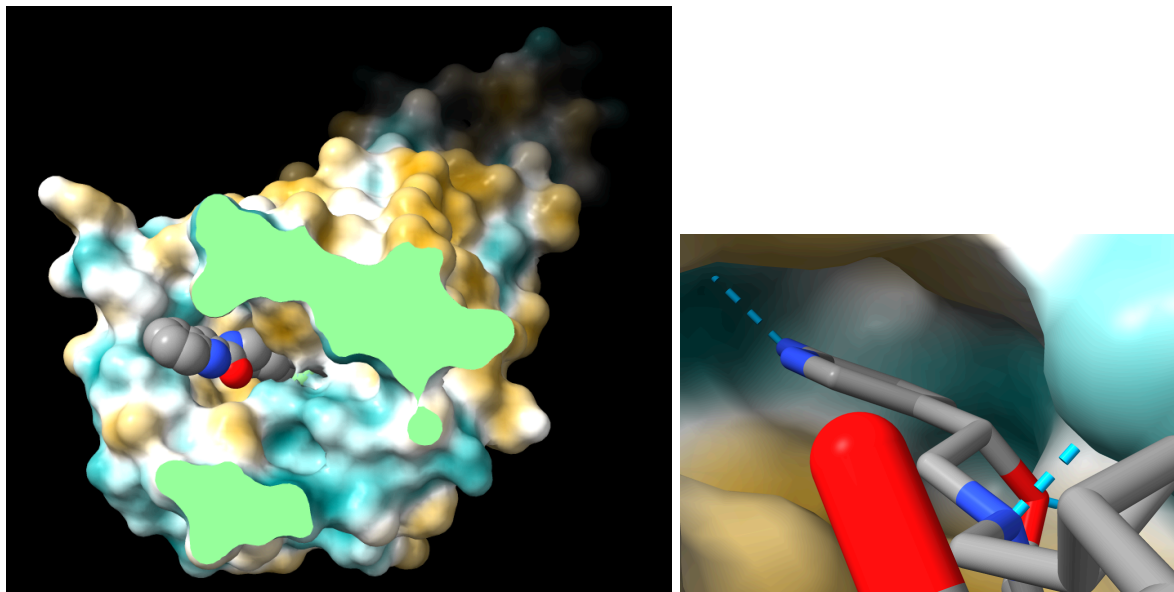


Fig 2. Extracellular view of active D3R with surface hydrophilicity coloring (blue: hydrophilic; orange: hydrophobic) and the bitopic ligand in atomic sphere form, alongside a closeup of hydrogen bonds between the bitopic ligand in stick representation and its binding sites.

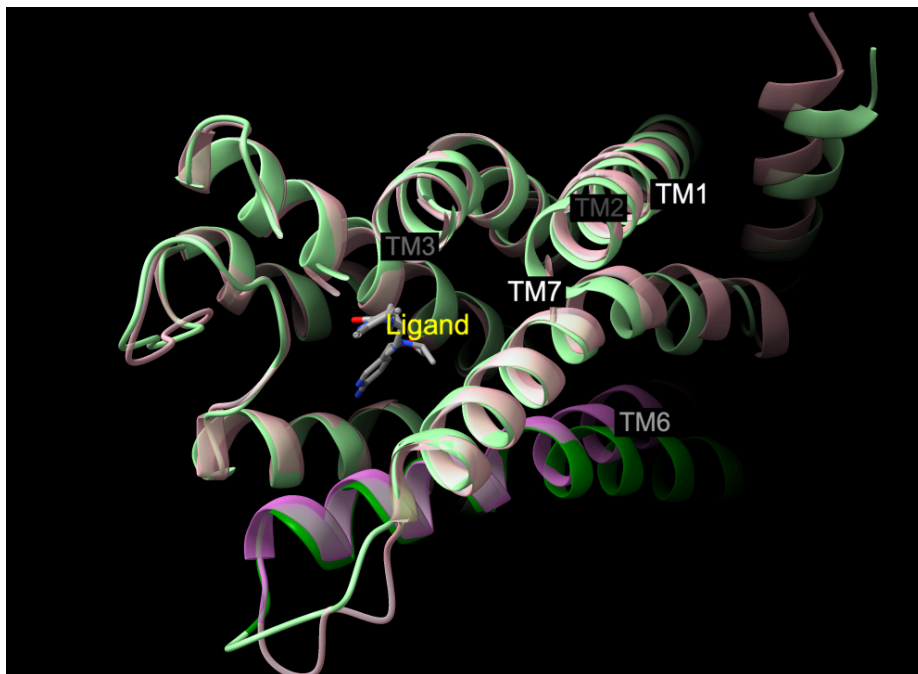


Fig 3. Extracellular view of the bitopic ligand alongside the D3 receptor inactive (pink) and active (green) conformations.

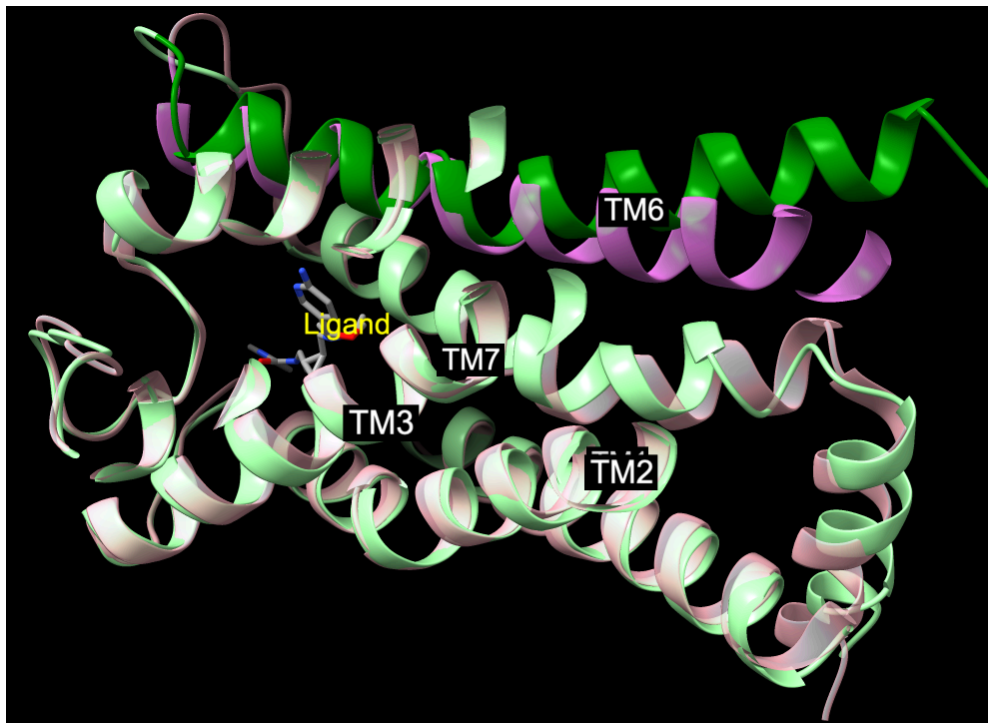


Fig 4. Side view of the bitopic ligand alongside the D3 inactive (pink) and active (green) receptors, better demonstrating the conformational changes upon FOB02-04A binding.

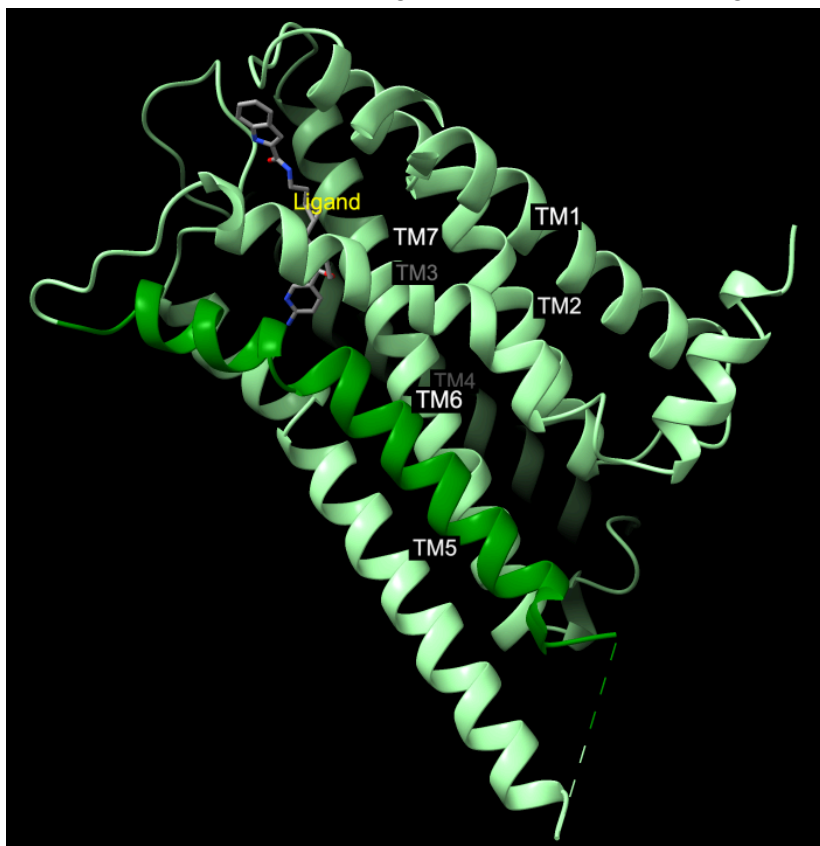


Fig 5. Side view of the bitopic ligand and the active D3 receptor, with TM6 visible in the front.

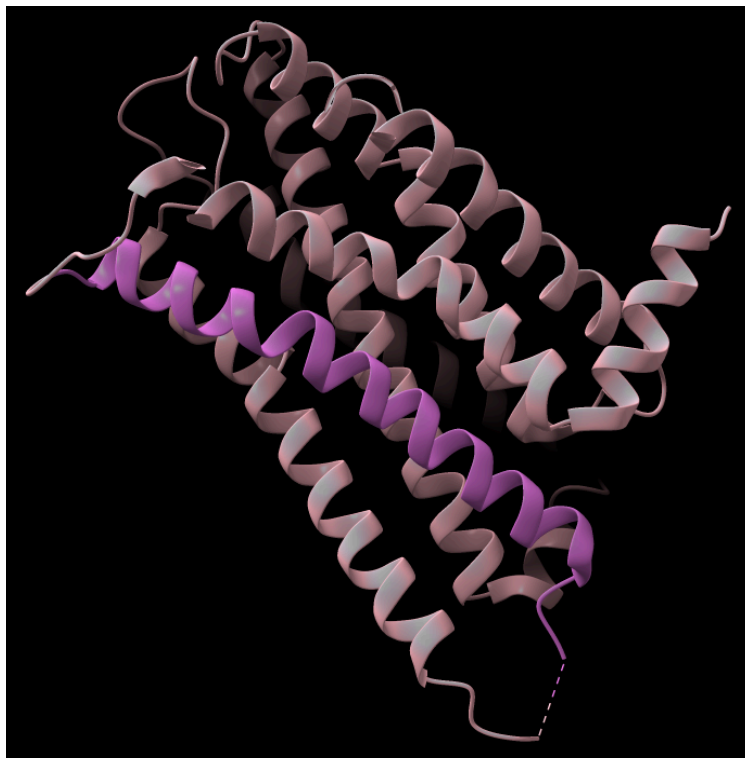


Fig 6. Side view of the inactive D3 receptor, with TM6 visible in the front.



Fig 7. Extracellular perspective of the D3R inactive conformation in ribbon form and bitopic ligand in ball stick form still H-bonding to visible residues of what would be the active conformation.

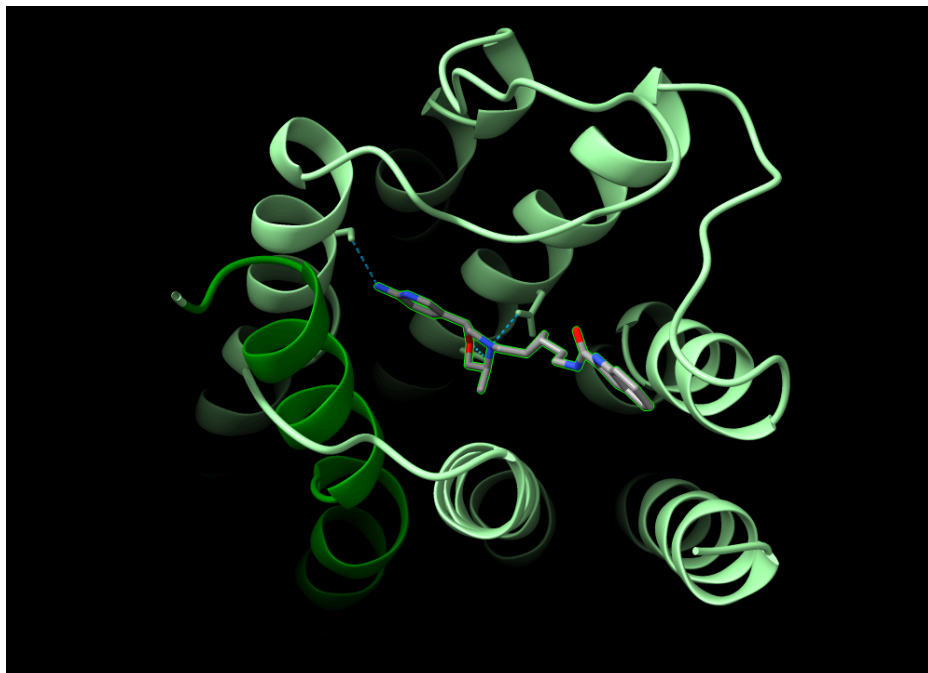


Fig 8. Extracellular perspective of the D3R active conformation in ribbon form and bitopic ligand in stick form.

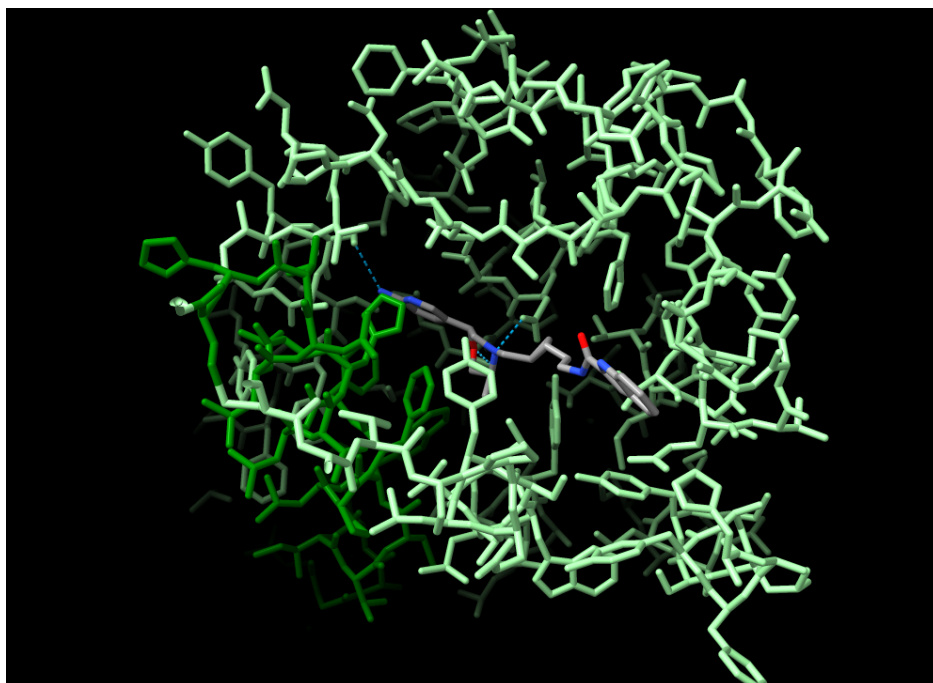


Fig 9. Extracellular perspective of the D3R active conformation and bitopic ligand both in stick form.

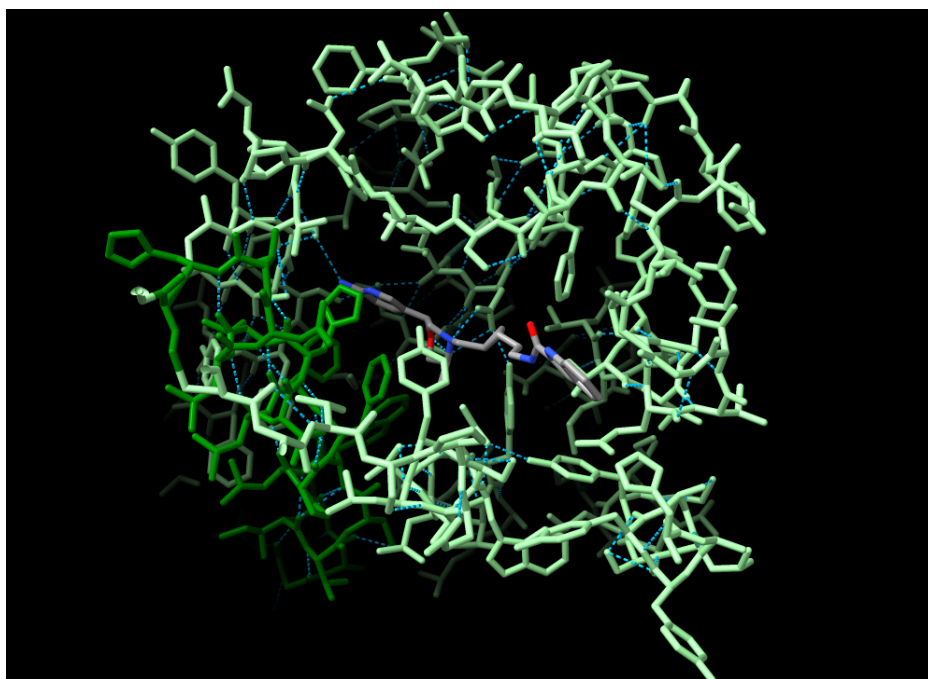


Fig 10. Extracellular perspective of the D3R active conformation and bitopic ligand both in stick form, with all hydrogen bonds visible.

First 3D Printed Model:

Onshape Document (Online 3D Modeling Software):

<https://cad.onshape.com/documents/9d71d3b84293a432f03b2224/w/c305c60de5291e26d04775f9/e/aa266522ae6e386aed240070?renderMode=0&uiState=69324c0d0d2694ebb8242581>

Video Demonstration:

<https://drive.google.com/file/d/18pIEZwxltCmTv3NBtCcqY7MytQIPH4oE/view?usp=sharing>

Images:

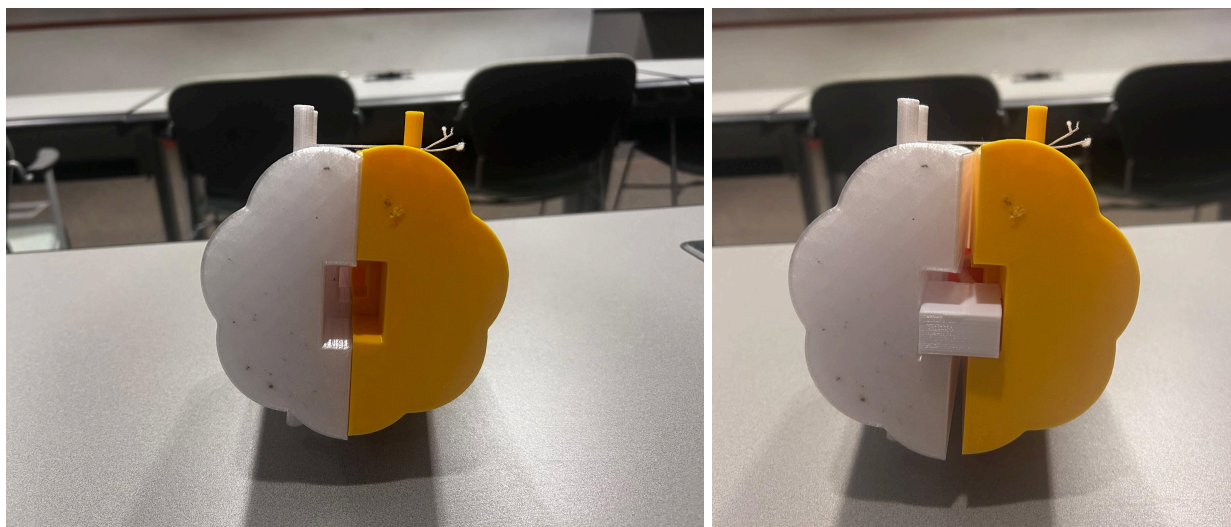


Fig 1. First 3D print model design depicting inactive vs. active conformations through insertion of bitopic ligand, if fit correctly, “opens” the two halves of the receptor to indicate its active positioning.

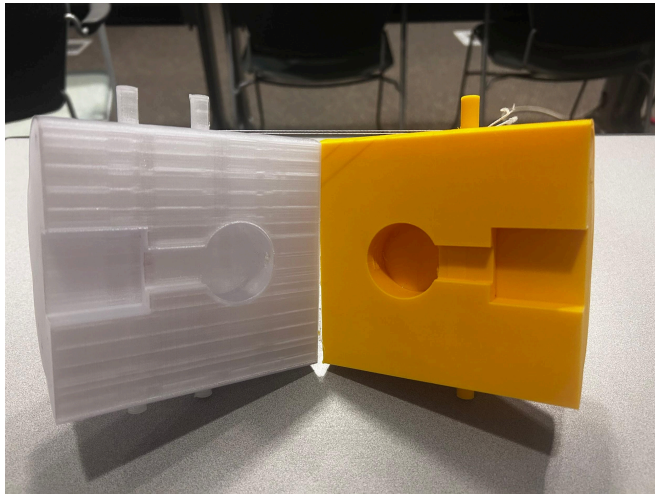


Fig 2. Receptor-only with insides visible.

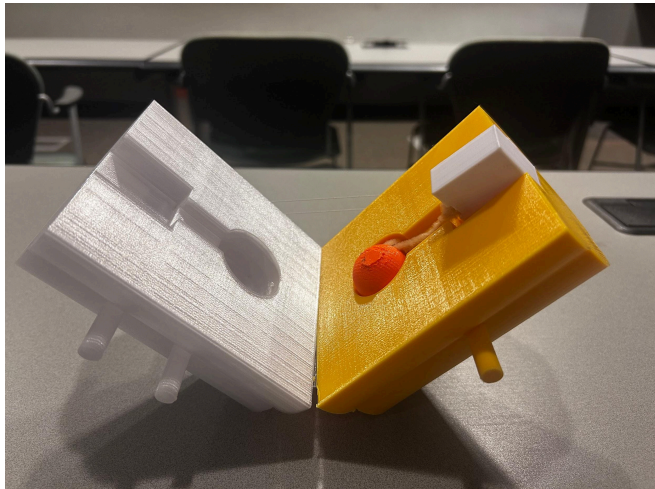


Fig 3. Ligand binding to orthosteric and allosteric pockets.



Fig 4. Side-view of D3 receptor with a bound bitopic ligand.

Final 3D Printed Model:

Onshape Document (Online 3D Modeling Software):

<https://cad.onshape.com/documents/faedf1b3e9cdaac4dac6b15e/w/441c22595c05f2daa720b04c/e/950ec2b1f99f807973e8783d?renderMode=0&uiState=69324a9ea308cf374b32576d>

Video of the model in its different conformations:

https://drive.google.com/file/d/1yZ_c30h6e0LN8_3JjLUjyJ_34UDwPcA9/view?usp=sharing

Video of the model interacting with the orthosteric and bitopic ligands:

https://drive.google.com/file/d/1C_DjaqikZIGduULoQ2THBfllO36VggCg/view?usp=sharing

Images:

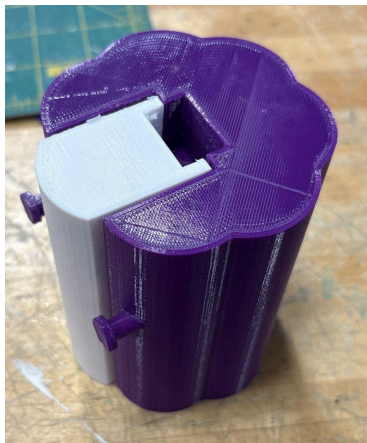


Image 1. Three-quarters view of the finished D3 receptor model

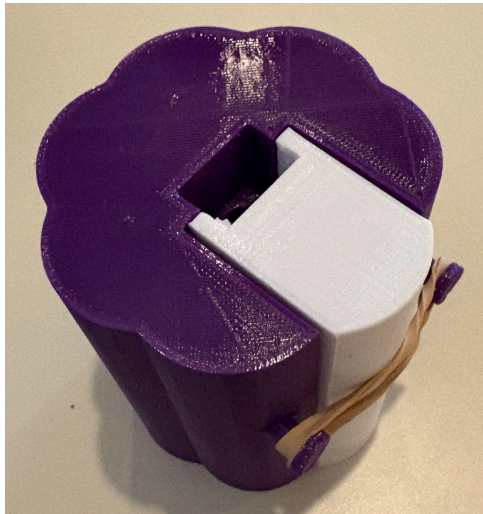


Image 2. Another three-quarters view of the finished D3 receptor model



Image 3. Side view of the D3 receptor



Image 4. Top view of the D3 receptor. The space in the middle show above is the orthosteric binding site.



Image 5. An inside look at the D3 receptor model. The round pocket is the allosteric binding site, while the rectangular pocket is the orthosteric binding site.



Image 6. A cardboard model of the orthosteric ligand, representing a traditional D3 receptor agonist such as dopamine



Image 7. A cardboard model of the bitopic ligand FOB02-04A, which has both allosteric and orthosteric pharmacophores. The allosteric pharmacophore is shown made of cardboard on the left side, while the cardboard orthosteric pharmacophore is on the right. The linker is made of pipe cleaners.